



59

Sardar Patel Institute of Technology

Bhavan's Campus, Munshi Nagar, Andheri (West), Mumbai-400058, India
(Autonomous College Affiliated to University of Mumbai)

End Semester Examination

25th April – 2018

Max. Marks: 100

Class: FYMCA

Course Code: MCA21

Name of the Course: Operating System

Duration: 3Hrs

Semester: II

Branch: MCA

Instruction:

- (1) All Questions are compulsory
- (2) Draw neat diagrams
- (3) Assume suitable data if necessary

Q No.		Max. Marks	CO																
Q.1 (a)	Illustrate various services provided by OS	05	CO1																
(b)	Classify different types of OS based on their functionality	05	CO1																
(c)	Illustrate the concept of demand paging	05	CO3																
(d)	How does logical address is translated in to physical address in paging?	05	CO3																
Q.2 (a)	Consider the system with 3 processes P1, P2, P3. Analyze whether the given processes are schedulable or not using Rate Monotonic Scheduling Algorithm (1 is the highest Priority)	05	CO2																
	<table><tr><th>Process Name</th><th>Service Time</th><th>Time Period(Repeat Frequency)</th><th>Priority</th></tr><tr><td>P1</td><td>1</td><td>4</td><td>1</td></tr><tr><td>P2</td><td>2</td><td>5</td><td>2</td></tr><tr><td>P3</td><td>2</td><td>7</td><td>3</td></tr></table>	Process Name	Service Time	Time Period(Repeat Frequency)	Priority	P1	1	4	1	P2	2	5	2	P3	2	7	3		
Process Name	Service Time	Time Period(Repeat Frequency)	Priority																
P1	1	4	1																
P2	2	5	2																
P3	2	7	3																
(b)	Differentiate between user level threads and kernel level thread(at least 5 points)	05	CO2																

(c)

Assume you have the following jobs to execute with one processor, with the jobs arriving in the order listed here:

Process Name	Arrival Time	Burst Time
P0	0	250
P1	50	170
P2	130	75
P3	190	100
P4	210	130
P5	350	50

Use the following CPU Scheduling algorithm to calculate Turnaround Time (TAT) and Waiting Time (WT)

- RR with Time Quantum=100
- FCFS
- SJF(preemptive)

04
03
03

CO2

OR

Q.2 (a) Demonstrate process state diagram

(b) Compare different types of schedulers

05

CO2

(c) 1. Consider below table of four processes under multilevel queue scheduling. Queue number denotes the queue of the process.

05

CO2

Process Name	Arrival Time	Burst Time	Queue No
P1	0	4	1
P2	0	3	1
P3	0	8	2
P4	10	5	1

Priority of queue 1 is greater than queue 2. Queue 1 uses Round Robin (Time Quantum = 2) and queue 2 uses FCFS.

Draw Gantt chart for the processes given above and calculate waiting time for process P3

2. Compare Multilevel queue scheduling with multilevel feedback queue scheduling

05

CO2

Q.3 (a)

There is a shared resource which should be accessed by multiple processes. There are two types of processes in this context. They are reader and writer. Any number of readers can read from the shared resource simultaneously, but only one writer can write to the shared resource. When a writer is writing data to the resource, no other process can access the resource. A writer cannot write to the resource if there are non-zero number of readers accessing the resource at that time. How does this problem can be solved using semaphores

10

CO2

OR

Define Semaphore

01

CO2

State how does semaphores solve problem of

1. Critical section

03

2. Order of execution among the processes(Give example)

03

3. Resource management(Give example)

03

(b)

1. Consider the following snapshot of system

CO2

	Allocation				Max				Available			
	A	B	C	D	A	B	C	D	A	B	C	D
P0	0	1	1	0	0	2	1	0	1	5	2	0
P1	1	2	3	1	1	6	5	2				
P2	1	3	6	5	2	3	6	6				
P3	0	6	3	2	0	6	5	2				
P4	0	0	1	4	0	6	5	6				

Using Banker's Algorithm

i. How many resources of type A, B, C, D are there?

01

ii. What is the content of Need Matrix

02

iii. Is the system in safe state? If so, Give safe sequence.

03

2. Describe the necessary conditions for deadlock to occur.

04

CO2

Q.4 (a)

Consider the current scenario of memory

CO3

200	400	600	500	300	250

There are 4 processes in system with following requests P1=357K, P2=210K, P3=468K, P4=491K Apply Following Allocation scheme For Fixed Size Partitioning scheme

1. First Fit

01

2. Best Fit

01

3. Worst Fit

01

4. Calculate total space get wasted due to internal fragmentation for Best fit and Worst Fit

02

(b)

Compare Paging and Segmentation

05

CO3

(c)

Define Page faults? Consider the following page reference string

01

CO3

a, b, c, d, c, a, d, b, e, b, a, b, c, d

Apply following page replacement algorithms

1. FIFO

03

2. LRU

03

3. Optimal

03

How many page faults would occur assuming 4 frames? Initially all frames are empty

	OR Suppose that the disk drive has 200 cylinders, numbered 0-199. The drive is currently servicing the request at cylinder 100. and the previous request was at 130. The pending requests are 55, 58, 39, 18, 90, 160, 150, 38, 84 What is the total distance of cylinders that the disk arm moves to satisfy all pending requests for each of the following disk scheduling algorithms? 1. FCFS 2. SCAN 3. C-SCAN 4. LOOK	03 03 02 02	CO3
Q.5 (a)	Discuss and Compare different File Allocation Methods (based on their technique, advantages, and disadvantages)	06	CO4
(b)	1. Suppose a disk has 32M blocks, each block of 2Kbytes, how many blocks are needed for managing a bit-vector? 2. What is the purpose of a file control block (FCB)?	02 02	CO4
(c)	1. Define Protection. Illustrate the concept of Access Matrix with the help of an example 2. Compare the different RAID Levels (0, 1, 5, 6, and 10) with suitable diagram. List out the advantage and disadvantage of each	05 05	CO4 CO4